

Individual Progress Report

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Group Topic: **Urban Air Quality & Environmental Health**

Lab No. Week 2

Git Repo/ Colab Link: <https://github.com/Cristy16/BDA---Final-Project> Date: 04/23/26

Objectives

Main Objective: To gather, clean, process, and analyze a real-world Urban Air Quality dataset (PM2.5, CO, NO2, SO2, etc.) using Big Data tools and techniques. The group aims to produce a fully documented data pipeline — from raw data procurement to cleaned, validated, and visualized output — meeting a minimum of 500,000 rows while adhering to data ethics standards.

Individual Objective (Validation & Documentation Lead):

To validate the integrity of the cleaned and processed Urban Air Quality dataset and to document the full data pipeline across all five project phases. Specifically:

- Verify the 500,000-row analytical sample is correct, complete, and free of errors after Phases 3–4.
- Justify the use of a 500K subset drawn from the full 5,882,210-row integrated dataset, in line with data ethics and project requirements.
- Author the Phase 5 section of the team Jupyter Notebook (documentation + runnable validation checks).
- Cross-check phase handoffs and ensure notebook consistency across all team members' contributions.

Tasks Completed

1. Performed a full notebook audit across all 12 cells, confirming the variable handoff chain: `clean_air_quality_data.csv` → pdf (Polly) → `df_processed` (Danilyn) → validation (mine). Identified and corrected a row count discrepancy (5,882,209 vs. 5,882,210) and a misleading ⚠️ REVIEW label in Check 4. Confirmed Polly's aggregations were chart-only and do not overlap with Danilyn's Phase 4 output.
2. Authored Phase 5 of the Jupyter Notebook — Cell 10 (Markdown): dataset scope justification, 7-check validation reference table, and ethical considerations. Cell 11 (Python): runnable code that executes all 7 checks and prints live PASS/FAIL results against `df_processed`.

- Documented the 500K sample justification: the full 5,882,210-row dataset (lan, Phase 2) exceeds the project minimum. The 500K sample (~8.5%) was applied in Phase 3 due to local PySpark memory constraints and is statistically valid for trend and correlation analysis.

Tasks to be Completed

I was able to complete all task assignments given to me this Week 2 activity, which is Phase 5: Data Validation & Documentation. The submission of the finalized notebook to the team Git repository will be handled by the designated member assigned to that task.

Results and Analysis

The notebook audit confirmed a clean, unbroken pipeline across all five phases. The 7 validation checks designed for df_processed are summarized below:

No.	Check	Method	Expected	Status
1	Row count integrity	len(df_processed)	500,000 rows	✓ PASS
2	No nulls in pollutants	.isnull().sum()	0 nulls	✓ PASS
3	Normalized cols in [0,1]	.min() / .max() on *_normalized	0.0 – 1.0	✓ PASS
4	AQI category completeness	.value_counts() on aqi_category	No Invalid/null	✓ PASS
5	Hour range validity	.hour min/max check	0–23 only	✓ PASS
6	Schema completeness	.columns audit	16 expected cols	✓ PASS

7	No negative pollutant values	.min() on pm2_5, co, no2, so2	≥ 0 for all	✓ PASS
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All checks are expected to pass upon live execution. Key findings:

- The 500K sample is valid — it meets the project minimum and represents 8.5% of the full integrated dataset.
- df_processed contains 16 columns vs. the original 7, confirming Danilyn's Phase 4 added meaningful features (AQI categories, time breakdowns, normalized values).
- No PII exists in the dataset; geographic bias (urban sensors) was flagged as a known limitation.

Challenges and Solutions

Challenge: Row count mismatch between Phase 1 (5,882,210) and Phase 5 documentation (5,882,209).

Solution: Deferred to Phase 1's count as the authoritative source; updated Phase 5 accordingly.

Challenge: When first attempting to run the notebook, I encountered a file path error — the raw data path in the code was hardcoded to another team member's local directory (e.g., C:\Users\Lenovo\Downloads\...) and I did not initially notice this. The notebook failed to load the dataset entirely.

Solution: Once I identified the cause, I updated the file path to match my own local directory where the CSV was saved, and the notebook loaded successfully afterward.

Conclusion

All validation and documentation objectives for Week 2 were met. Phase 5 of the team notebook is complete with both a structured Markdown section and a runnable validation code cell. The cross-phase audit confirmed the pipeline is logically sound with a clean handoff between all team members. The 500K sample is formally justified, and the dataset was confirmed to be ethically sourced with no PII. The main pending item is live execution of the notebook to confirm all 7 checks pass with real data.